JEE MAIN · MATHEMATICS · CHAPTER-WISE PYQS

Area Under The Curves

JEE Main Mathematics · Previous Year Questions with Answer Key

170	160	2002 – 2026	English
QUESTIONS	PAPERS COVERED	YEARS	LANGUAGE

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MCQ (Single Correct Answer)	124 questions
Numerical	46 questions
Answer key	at the end

1. **Solve**
   Print it, or work through it on screen.
2. **Check**
   The answer key is on the last pages.

MCQ (Single Correct Answer) 124 questions

Q1. JEE Main 2026 (Online) 8th April Evening Shift

Let $f:(1,\infty)\to\mathbf{R}$ be a function defined as $f(x)=\frac{x-1}{x+1}$. Let $f^{i+1}(x)=f\left(f^{i}(x)\right), i=1,2,\ldots,25$, where $f^{1}(x)=f(x)$. If $g(x)+f^{26}(x)=0, x\in(1,\infty)$, then the area of the region bounded by the curves $y=g(x), 2y=2x-3, y=0$ and $x=4$ is :

(A) $\frac{1}{8}+\log_{\mathrm{e}}2$ (B) $\frac{1}{4}+\log_{\mathrm{e}}2$ (C) $\frac{5}{6}+3\log_{e}2$ (D) $\frac{5}{6}+\log_{e}2$

Q2. JEE Main 2026 (Online) 6th April Evening Shift

The area of the region $\left\{(x,y):x^2-8x\le y\le -x\right\}$ is :

(A) $\frac{343}{6}$ (B) $\frac{637}{6}$ (C) $\frac{437}{6}$ (D) $\frac{523}{6}$

Q3. JEE Main 2026 (Online) 6th April Morning Shift

The area of the region $\left\{(x,y):0\le y\le 6-x, y^2\ge 4x-3, x\ge 0\right\}$ is :

(A) 8 (B) 9 (C) 12 (D) 15

Q4. JEE Main 2026 (Online) 6th April Morning Shift

Let e be the base of natural logarithm and let $f:\{1,2,3,4\}\to\left\{1,e,e^2,e^3\right\}$ and $g:\left\{1,e,e^2,e^3\right\}\to\left\{1,\frac{1}{2},\frac{1}{3},\frac{1}{4}\right\}$ be two bijective functions such that f is strictly decreasing and g is strictly increasing. If $\phi(x)=\left[f^{-1}\left\{g^{-1}\left(\frac{1}{2}\right)\right\}\right]^{x}$, then the area of the region $\mathrm{R}=\left\{(x,y):x^2\le y\le\phi(x),0\le x\le 1\right\}$ is :

(A) $\frac{3-\log_e(2)}{3\log_e(2)}$ (B) $\frac{1}{3\log_e(2)}$

(C) $3+\log_e(2)$ (D) $\frac{3+\log_e(2)}{2+\log_e(3)}$

Q5. JEE Main 2026 (Online) 5th April Morning Shift

The area of the region $\mathrm{R}=\left\{(x,y):xy\le 27, 1\le y\le x^2\right\}$ is equal to :

(A) $78\log_e 3-\frac{52}{3}$ (B) $54\log_e 3-\frac{52}{3}$ (C) $54\log_e 3-\frac{26}{3}$ (D) $54\log_e 3+\frac{26}{3}$

Q6. JEE Main 2026 (Online) 4th April Evening Shift

The area of the region bounded by the curves $x+3y^2=0$ and $x+4y^2=1$ is equal to :

(A) $\frac{1}{3}$ (B) $\frac{2}{3}$ (C) $\frac{4}{3}$ (D) $\frac{5}{3}$

Q7. JEE Main 2026 (Online) 4th April Morning Shift

The area of the region $\{(x, y) : y \leq \pi - |x|, y \leq |x \sin x|, y \geq 0\}$ is:

(A) $1 + \frac{\pi^2}{8}$ (B) $2 + \frac{\pi^2}{4}$ (C) $\frac{\pi^2}{8} - 1$ (D) $4 + \frac{\pi^2}{2}$

Q8. JEE Main 2026 (Online) 28th January Evening Shift

Let $P_1 : y = 4x^2$ and $P_2 : y = x^2 + 27$ be two parabolas. If the area of the bounded region enclosed between P_1 and P_2 is six times the area of the bounded region enclosed between the line $y = \alpha x$, $\alpha > 0$ and P_1, then α is equal to :

(A) 12 (B) 15 (C) 8 (D) 6

Q9. JEE Main 2026 (Online) 28th January Morning Shift

The area of the region $\mathrm{R} = \{(x, y) : xy \leq 8, 1 \leq y \leq x^2, x \geq 0\}$ is

(A) $\frac{2}{3}(20 \log_e(2) + 9)$ (B) $\frac{1}{3}(40 \log_e(2) + 27)$

(C) $\frac{1}{3}(49 \log_e(2) - 15)$ (D) $\frac{2}{3}(24 \log_e(2) - 7)$

Q10. JEE Main 2026 (Online) 24th January Evening Shift

Let $f(\alpha)$ denote the area of the region in the first quadrant bounded by $x = 0, x = 1, y^2 = x$ and $y = |\alpha x - 5| - |1 - \alpha x| + \alpha x^2$. Then $(f(0) + f(1))$ is equal to

(A) 12 (B) 14 (C) 9 (D) 7

Q11. JEE Main 2026 (Online) 24th January Morning Shift

Let A_1 be the bounded area enclosed by the curves $y = x^2 + 2, x + y = 8$ and y-axis that lies in the first quadrant. Let A_2 be the bounded area enclosed by the curves $y = x^2 + 2, y^2 = x, x = 2$, and y-axis that lies in the first quadrant. Then $\mathrm{A}_1 - \mathrm{A}_2$ is equal to

(A) $\frac{2}{3}(2\sqrt{2} + 1)$ (B) $\frac{2}{3}(3\sqrt{2} + 1)$ (C) $\frac{2}{3}(\sqrt{2} + 1)$ (D) $\frac{2}{3}(4\sqrt{2} + 1)$

Q12. JEE Main 2026 (Online) 23rd January Evening Shift

The area of the region enclosed between the circles $x^2 + y^2 = 4$ and $x^2 + (y - 2)^2 = 4$ is:

(A) $\frac{2}{3}(4\pi - 3\sqrt{3})$ (B) $\frac{4}{3}(2\pi - \sqrt{3})$ (C) $\frac{4}{3}(2\pi - 3\sqrt{3})$ (D) $\frac{2}{3}(2\pi - 3\sqrt{3})$

Q13. JEE Main 2026 (Online) 22nd January Evening Shift

The area of the region $\mathrm{A} = \{(x, y) : 4x^2 + y^2 \leqslant 8 \text{ and } y^2 \leqslant 4x\}$ is:

(A) $\pi + \frac{2}{3}$ (B) $\frac{\pi}{2} + 2$ (C) $\pi + 4$ (D) $\frac{\pi}{2} + \frac{1}{3}$

Q14. JEE Main 2026 (Online) 22nd January Morning Shift

Let the line $x = -1$ divide the area of the region $\{(x, y) : 1 + x^2 \leq y \leq 3 - x\}$ in the ratio $m : n, \gcd(m, n) = 1$. Then $m + n$ is equal to

(A) 27 (B) 28 (C) 25 (D) 26

Q15. JEE Main 2026 (Online) 21st January Evening Shift

If the area of the region $\{(x, y) : 1 - 2x \leq y \leq 4 - x^2,\ x \geq 0,\ y \geq 0\}$ is $\frac{\alpha}{\beta}$, $\alpha, \beta \in \mathbb{N}, \gcd(\alpha, \beta) = 1$, then the value of $(\alpha + \beta)$ is:

(A) 73 (B) 85 (C) 67 (D) 91

Q16. JEE Main 2026 (Online) 21st January Morning Shift

The area of the region, inside the ellipse $x^2 + 4y^2 = 4$ and outside the region bounded by the curves $y = |x| - 1$ and $y = 1 - |x|$, is :

(A) $2\pi - 1$ (B) $3(\pi - 1)$ (C) $2(\pi - 1)$ (D) $2\pi - \frac{1}{2}$

Q17. JEE Main 2025 (Online) 7th April Evening Shift

If the area of the region $\{(x, y) : 1 + x^2 \leq y \leq \min\{x + 7, 11 - 3x\}\}$ is A, then $3A$ is equal to :

(A) 50 (B) 46 (C) 49 (D) 47

Q18. JEE Main 2025 (Online) 7th April Morning Shift

If the area of the region bounded by the curves $y = 4 - \frac{x^2}{4}$ and $y = \frac{x-4}{2}$ is equal to α, then 6α. equals

(A) 210 (B) 250 (C) 240 (D) 220

Q19. JEE Main 2025 (Online) 4th April Morning Shift

Let $f : [0, \infty) \to \mathbb{R}$ be a differentiable function such that

$f(x) = 1 - 2x + \int_0^x e^{x-t} f(t)dt$ for all $x \in [0, \infty)$.

Then the area of the region bounded by $y = f(x)$ and the coordinate axes is

(A) $\sqrt{5}$ (B) 2 (C) $\sqrt{2}$ (D) $\frac{1}{2}$

Q20. JEE Main 2025 (Online) 3rd April Evening Shift

The area of the region $\{(x, y) : |x - y| \leq y \leq 4\sqrt{x}\}$ is

(A) $\frac{512}{3}$ (B) $\frac{2048}{3}$ (C) 512 (D) $\frac{1024}{3}$

Q21. JEE Main 2025 (Online) 29th January Evening Shift

Let the area enclosed between the curves $|y| = 1 - x^2$ and $x^2 + y^2 = 1$ be α. If $9\alpha = \beta\pi + \gamma; \beta, \gamma$ are integers, then the value of $|\beta - \gamma|$ equals:

(A) 15 (B) 18 (C) 33 (D) 27

Q22. JEE Main 2025 (Online) 29th January Morning Shift

Let the area of the region

$(x, y) : 2y \leq x^2 + 3,\ y + |x| \leq 3,\ y \geq |x - 1|$ be A. Then $6A$ is equal to :

(A) 14 (B) 18 (C) 16 (D) 12

Q23. JEE Main 2025 (Online) 28th January Evening Shift

The area of the region bounded by the curves $x(1 + y^2) = 1$ and $y^2 = 2x$ is:

(A) $\frac{\pi}{4} - \frac{1}{3}$ (B) $\frac{\pi}{2} - \frac{1}{3}$

(C) $2\left(\frac{\pi}{2} - \frac{1}{3}\right)$ (D) $\frac{1}{2}\left(\frac{\pi}{2} - \frac{1}{3}\right)$

Q24. JEE Main 2025 (Online) 28th January Morning Shift

The area (in sq. units) of the region $\left\{(x, \mathrm{y}) : 0 \leq \mathrm{y} \leq 2|x| + 1, 0 \leq \mathrm{y} \leq x^2 + 1, |x| \leq 3\right\}$ is

(A) $\frac{32}{3}$ (B) $\frac{64}{3}$ (C) $\frac{17}{3}$ (D) $\frac{80}{3}$

Q25. JEE Main 2025 (Online) 24th January Evening Shift

The area of the region enclosed by the curves $y = \mathrm{e}^x, y = |\mathrm{e}^x - 1|$ and y-axis is :

(A) $1 + \log_{\mathrm{e}} 2$ (B) $\log_{\mathrm{e}} 2$ (C) $1 - \log_{\mathrm{e}} 2$ (D) $2\log_{\mathrm{e}} 2 - 1$

Q26. JEE Main 2025 (Online) 24th January Morning Shift

The area of the region $\left\{(x, y) : x^2 + 4x + 2 \leq y \leq |x + 2|\right\}$ is equal to

(A) 7 (B) 24/5 (C) 20/3 (D) 5

Q27. JEE Main 2025 (Online) 23rd January Evening Shift

If the area of the region $\left\{(x, y) : -1 \leq x \leq 1, 0 \leq y \leq \mathrm{a} + \mathrm{e}^{|x|} - \mathrm{e}^{-x}, \mathrm{a} > 0\right\}$ is $\frac{\mathrm{e}^2 + 8\mathrm{e} + 1}{\mathrm{e}}$, then the value of a is :

(A) 7 (B) 5 (C) 6 (D) 8

Q28. JEE Main 2025 (Online) 22nd January Evening Shift

The area of the region enclosed by the curves $y = x^2 - 4x + 4$ and $y^2 = 16 - 8x$ is :

(A) $\frac{8}{3}$ (B) 5 (C) 8 (D) $\frac{4}{3}$

Q29. JEE Main 2025 (Online) 22nd January Morning Shift

The area of the region, inside the circle $(x - 2\sqrt{3})^2 + y^2 = 12$ and outside the parabola $y^2 = 2\sqrt{3}x$ is :

(A) $3\pi - 8$ (B) $6\pi - 8$ (C) $3\pi + 8$ (D) $6\pi - 16$

Q30. JEE Main 2024 (Online) 9th April Evening Shift

The area (in square units) of the region enclosed by the ellipse $x^2 + 3y^2 = 18$ in the first quadrant below the line $y = x$ is

(A) $\sqrt{3}\pi + 1$ (B) $\sqrt{3}\pi$ (C) $\sqrt{3}\pi - \frac{3}{4}$ (D) $\sqrt{3}\pi + \frac{3}{4}$

Q31. JEE Main 2024 (Online) 9th April Morning Shift

The parabola $y^2 = 4x$ divides the area of the circle $x^2 + y^2 = 5$ in two parts. The area of the smaller part is equal to :

(A) $\frac{2}{3} + 5\sin^{-1}\left(\frac{2}{\sqrt{5}}\right)$ (B) $\frac{2}{3} + \sqrt{5}\sin^{-1}\left(\frac{2}{\sqrt{5}}\right)$

(C) $\frac{1}{3} + 5\sin^{-1}\left(\frac{2}{\sqrt{5}}\right)$ (D) $\frac{1}{3} + \sqrt{5}\sin^{-1}\left(\frac{2}{\sqrt{5}}\right)$

Q32. JEE Main 2024 (Online) 8th April Evening Shift

The area of the region in the first quadrant inside the circle $x^2 + y^2 = 8$ and outside the parabola $y^2 = 2x$ is equal to :

(A) $\frac{\pi}{2} - \frac{1}{3}$ (B) $\pi - \frac{1}{3}$ (C) $\pi - \frac{2}{3}$ (D) $\frac{\pi}{2} - \frac{2}{3}$

Q33. JEE Main 2024 (Online) 6th April Evening Shift

If the area of the region $\left\{(x, y) : \frac{a}{x^2} \leq y \leq \frac{1}{x}, 1 \leq x \leq 2, 0 < a < 1\right\}$ is $(\log_e 2) - \frac{1}{7}$ then the value of $7a - 3$ is equal to :

(A) 1 (B) 0 (C) 2 (D) -1

Q34. JEE Main 2024 (Online) 6th April Morning Shift

Let the area of the region enclosed by the curves $y = 3x, 2y = 27 - 3x$ and $y = 3x - x\sqrt{x}$ be A. Then $10A$ is equal to

(A) 172 (B) 154 (C) 162 (D) 184

Q35. JEE Main 2024 (Online) 5th April Evening Shift

The area enclosed between the curves $y = x|x|$ and $y = x - |x|$ is :

(A) $\frac{8}{3}$ (B) $\frac{2}{3}$ (C) $\frac{4}{3}$ (D) 1

Q36. JEE Main 2024 (Online) 4th April Evening Shift

The area (in sq. units) of the region described by $\{(x, y) : y^2 \leq 2x, \text{ and } y \geq 4x - 1\}$ is

(A) $\frac{9}{32}$ (B) $\frac{11}{12}$ (C) $\frac{8}{9}$ (D) $\frac{11}{32}$

Q37. JEE Main 2024 (Online) 4th April Morning Shift

One of the points of intersection of the curves $y = 1 + 3x - 2x^2$ and $y = \frac{1}{x}$ is $\left(\frac{1}{2}, 2\right)$. Let the area of the region enclosed by these curves be $\frac{1}{24}(l\sqrt{5} + \mathrm{m}) - \mathrm{n}\log_e(1 + \sqrt{5})$, where $l,\ \mathrm{m}, \mathrm{n} \in \mathbf{N}$. Then $l + \mathrm{m} + \mathrm{n}$ is equal to

(A) 30 (B) 29 (C) 31 (D) 32

Q38. JEE Main 2024 (Online) 1st February Morning Shift

The area enclosed by the curves $xy + 4y = 16$ and $x + y = 6$ is equal to :

(A) $28 - 30\log_e 2$ (B) $30 - 28\log_e 2$ (C) $30 - 32\log_e 2$ (D) $32 - 30\log_e 2$

Q39. JEE Main 2024 (Online) 31st January Evening Shift

The area of the region enclosed by the parabolas $y = 4x - x^2$ and $3y = (x - 4)^2$ is equal to :

(A) $\frac{32}{9}$ (B) $\frac{14}{3}$ (C) 4 (D) 6

Q40. JEE Main 2024 (Online) 31st January Morning Shift

The area of the region $\left\{(x, y) : y^2 \leq 4x, x < 4, \frac{xy(x-1)(x-2)}{(x-3)(x-4)} > 0, x \neq 3\right\}$ is

(A) $\frac{32}{3}$ (B) $\frac{16}{3}$ (C) $\frac{8}{3}$ (D) $\frac{64}{3}$

Q41. JEE Main 2024 (Online) 30th January Morning Shift

The area (in square units) of the region bounded by the parabola $y^2 = 4(x - 2)$ and the line $y = 2x - 8$, is :

(A) 7 (B) 8 (C) 9 (D) 6

Q42. JEE Main 2023 (Online) 13th April Evening Shift

The area of the region $\{(x, y) : x^2 \leq y \leq |x^2 - 4|, y \geq 1\}$ is

(A) $\frac{4}{3}(4\sqrt{2} + 1)$ (B) $\frac{3}{4}(4\sqrt{2} + 1)$ (C) $\frac{4}{3}(4\sqrt{2} - 1)$ (D) $\frac{3}{4}(4\sqrt{2} - 1)$

Q43. JEE Main 2023 (Online) 13th April Morning Shift

The area of the region enclosed by the curve $f(x) = \max\{\sin x, \cos x\}, -\pi \leq x \leq \pi$ and the x-axis is

(A) $2\sqrt{2}(\sqrt{2} + 1)$ (B) 4 (C) $2(\sqrt{2} + 1)$ (D) $4(\sqrt{2})$

Q44. JEE Main 2023 (Online) 12th April Morning Shift

The area of the region enclosed by the curve $y = x^3$ and its tangent at the point $(-1, -1)$ is :

(A) $\frac{23}{4}$ (B) $\frac{19}{4}$ (C) $\frac{27}{4}$ (D) $\frac{31}{4}$

Q45. JEE Main 2023 (Online) 11th April Morning Shift

Area of the region $\{(x, y) : x^2 + (y - 2)^2 \leq 4, x^2 \geq 2y\}$ is

(A) $2\pi + \frac{16}{3}$ (B) $\pi - \frac{8}{3}$ (C) $\pi + \frac{8}{3}$ (D) $2\pi - \frac{16}{3}$

Q46. JEE Main 2023 (Online) 8th April Morning Shift

The area of the region $\{(x, y) : x^2 \leq y \leq 8 - x^2, y \leq 7\}$ is :

(A) 18 (B) 24 (C) 20 (D) 21

Q47. JEE Main 2023 (Online) 6th April Evening Shift

The area bounded by the curves $y = |x - 1| + |x - 2|$ and $y = 3$ is equal to :

(A) 5 (B) 4 (C) 6 (D) 3

Q48. JEE Main 2023 (Online) 1st February Evening Shift

The area of the region given by $\{(x, y) : xy \leq 8, 1 \leq y \leq x^2\}$ is :

(A) $16\log_e 2 - \frac{14}{3}$ (B) $8\log_e 2 - \frac{13}{3}$ (C) $16\log_e 2 + \frac{7}{3}$ (D) $8\log_e 2 + \frac{7}{6}$

Q49. JEE Main 2023 (Online) 30th January Evening Shift

Let q be the maximum integral value of p in $[0, 10]$ for which the roots of the equation $x^2 - px + \frac{5}{4}p = 0$ are rational. Then the area of the region $\{(x, y) : 0 \leq y \leq (x - q)^2, 0 \leq x \leq q\}$ is :

(A) $\frac{125}{3}$ (B) 243 (C) 164 (D) 25

Q50. JEE Main 2023 (Online) 29th January Evening Shift

The area of the region $A = \{(x, y) : |\cos x - \sin x| \leq y \leq \sin x, 0 \leq x \leq \frac{\pi}{2}\}$ is

(A) $\sqrt{5} + 2\sqrt{2} - 4.5$

(B) $1 - \frac{3}{\sqrt{2}} + \frac{4}{\sqrt{5}}$

(C) $\sqrt{5} - 2\sqrt{2} + 1$

(D) $\frac{3}{\sqrt{5}} - \frac{3}{\sqrt{2}} + 1$

Q51. JEE Main 2023 (Online) 29th January Morning Shift

Let Δ be the area of the region $\{(x, y) \in R^2 : x^2 + y^2 \leq 21, y^2 \leq 4x, x \geq 1\}$. Then $\frac{1}{2}\left(\Delta - 21\sin^{-1}\frac{2}{\sqrt{7}}\right)$ is equal to

(A) $2\sqrt{3} - \frac{1}{3}$

(B) $2\sqrt{3} - \frac{2}{3}$

(C) $\sqrt{3} - \frac{4}{3}$

(D) $\sqrt{3} - \frac{2}{3}$

Q52. JEE Main 2023 (Online) 29th January Morning Shift

Let $A = \left\{(x, y) \in \mathbb{R}^2 : y \geq 0, 2x \leq y \leq \sqrt{4 - (x-1)^2}\right\}$ and

$B = \left\{(x, y) \in \mathbb{R} \times \mathbb{R} : 0 \leq y \leq \min\left\{2x, \sqrt{4 - (x-1)^2}\right\}\right\}$. .

Then the ratio of the area of A to the area of B is

(A) $\frac{\pi}{\pi+1}$

(B) $\frac{\pi-1}{\pi+1}$

(C) $\frac{\pi}{\pi-1}$

(D) $\frac{\pi+1}{\pi-1}$

Q53. JEE Main 2023 (Online) 25th January Evening Shift

Let T and C respectively be the transverse and conjugate axes of the hyperbola $16x^2 - y^2 + 64x + 4y + 44 = 0$. Then the area of the region above the parabola $x^2 = y + 4$, below the transverse axis T and on the right of the conjugate axis C is :

(A) $4\sqrt{6} - \frac{28}{3}$

(B) $4\sqrt{6} - \frac{44}{3}$

(C) $4\sqrt{6} + \frac{28}{3}$

(D) $4\sqrt{6} + \frac{44}{3}$

Q54. JEE Main 2023 (Online) 24th January Morning Shift

The area enclosed by the curves $y^2 + 4x = 4$ and $y - 2x = 2$ is :

(A) $\frac{22}{3}$

(B) 9

(C) $\frac{23}{3}$

(D) $\frac{25}{3}$

Q55. JEE Main 2022 (Online) 29th July Morning Shift

The area of the region

$\left\{(x, y) : |x - 1| \leq y \leq \sqrt{5 - x^2}\right\}$ is equal to :

(A) $\frac{5}{2}\sin^{-1}\left(\frac{3}{5}\right) - \frac{1}{2}$

(B) $\frac{5\pi}{4} - \frac{3}{2}$

(C) $\frac{3\pi}{4} + \frac{3}{2}$

(D) $\frac{5\pi}{4} - \frac{1}{2}$

Q56. JEE Main 2022 (Online) 28th July Evening Shift

The area enclosed by the curves $y = \log_e\left(x + e^2\right), x = \log_e\left(\frac{2}{y}\right)$ and $x = \log_e 2$, above the line $y = 1$ is:

(A) $2 + e - \log_e 2$ (B) $1 + e - \log_e 2$

(C) $e - \log_e 2$ (D) $1 + \log_e 2$

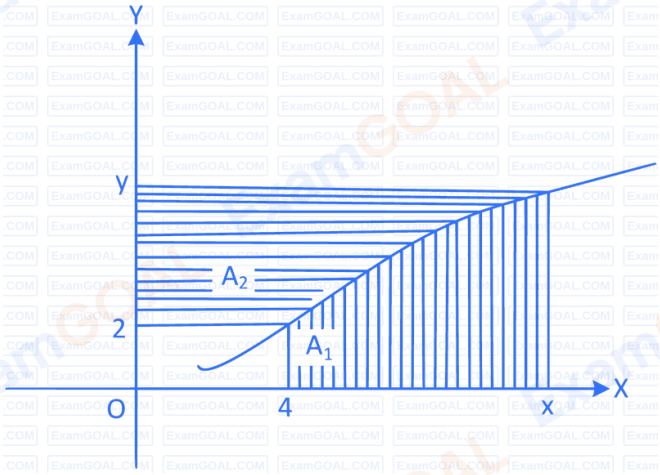
Q57. JEE Main 2022 (Online) 27th July Evening Shift

The area of the region enclosed by $y \leq 4x^2, x^2 \leq 9y$ and $y \leq 4$, is equal to :

(A) $\frac{40}{3}$ (B) $\frac{56}{3}$ (C) $\frac{112}{3}$ (D) $\frac{80}{3}$

Q58. JEE Main 2022 (Online) 27th July Evening Shift

Consider a curve $y = y(x)$ in the first quadrant as shown in the figure. Let the area A_1 is twice the area A_2. Then the normal to the curve perpendicular to the line $2x - 12y = 15$ does NOT pass through the point.



(A) (6, 21) (B) (8, 9) (C) (10, −4) (D) (12, −15)

Q59. JEE Main 2022 (Online) 27th July Morning Shift

The area of the smaller region enclosed by the curves $y^2 = 8x + 4$ and $x^2 + y^2 + 4\sqrt{3}x - 4 = 0$ is equal to

(A) $\frac{1}{3}(2 - 12\sqrt{3} + 8\pi)$ (B) $\frac{1}{3}(2 - 12\sqrt{3} + 6\pi)$ (C) $\frac{1}{3}(4 - 12\sqrt{3} + 8\pi)$ (D) $\frac{1}{3}(4 - 12\sqrt{3} + 6\pi)$

Q60. JEE Main 2022 (Online) 26th July Evening Shift

The area bounded by the curves $y = \left|x^2 - 1\right|$ and $y = 1$ is

(A) $\frac{2}{3}(\sqrt{2} + 1)$ (B) $\frac{4}{3}(\sqrt{2} - 1)$ (C) $2(\sqrt{2} - 1)$ (D) $\frac{8}{3}(\sqrt{2} - 1)$

Q61. JEE Main 2022 (Online) 26th July Morning Shift

The odd natural number a, such that the area of the region bounded by y = 1, y = 3, x = 0, x = y^a is $\frac{364}{3}$, is equal to :

(A) 3 (B) 5 (C) 7 (D) 9

Q62. JEE Main 2022 (Online) 25th July Morning Shift

The area of the region given by

$A = \{(x, y) : x^2 \leq y \leq \min\{x + 2, 4 - 3x\}\}$ is :

(A) $\frac{31}{8}$ (B) $\frac{17}{6}$ (C) $\frac{19}{6}$ (D) $\frac{27}{8}$

Q63. JEE Main 2022 (Online) 25th July Morning Shift

Let the locus of the centre $(\alpha, \beta), \beta > 0$, of the circle which touches the circle $x^2 + (y - 1)^2 = 1$ externally and also touches the x-axis be L. Then the area bounded by L and the line $y = 4$ is:

(A) $\frac{32\sqrt{2}}{3}$ (B) $\frac{40\sqrt{2}}{3}$ (C) $\frac{64}{3}$ (D) $\frac{32}{3}$

Q64. JEE Main 2022 (Online) 29th June Morning Shift

The area enclosed by $y^2 = 8x$ and $y = \sqrt{2}x$ that lies outside the triangle formed by $y = \sqrt{2}x$, $x = 1$, $y = 2\sqrt{2}$, is equal to:

(A) $\frac{16\sqrt{2}}{6}$ (B) $\frac{11\sqrt{2}}{6}$ (C) $\frac{13\sqrt{2}}{6}$ (D) $\frac{5\sqrt{2}}{6}$

Q65. JEE Main 2022 (Online) 28th June Evening Shift

The area of the bounded region enclosed by the curve

$y = 3 - \left|x - \frac{1}{2}\right| - |x + 1|$ and the x-axis is :

(A) $\frac{9}{4}$ (B) $\frac{45}{16}$ (C) $\frac{27}{8}$ (D) $\frac{63}{16}$

Q66. JEE Main 2022 (Online) 28th June Morning Shift

The area of the region S = {(x, y) : $y^2 \leq 8x$, $y \geq \sqrt{2}x$, $x \geq 1$} is

(A) $\frac{13\sqrt{2}}{6}$ (B) $\frac{11\sqrt{2}}{6}$ (C) $\frac{5\sqrt{2}}{6}$ (D) $\frac{19\sqrt{2}}{6}$

Q67. JEE Main 2022 (Online) 26th June Evening Shift

The area of the region bounded by $y^2 = 8x$ and $y^2 = 16(3 - x)$ is equal to:

(A) $\frac{32}{3}$ (B) $\frac{40}{3}$ (C) 16 (D) 19

Q68. JEE Main 2022 (Online) 26th June Morning Shift

The area bounded by the curve $y = |x^2 - 9|$ and the line $y = 3$ is :

(A) $4(2\sqrt{3} + \sqrt{6} - 4)$ (B) $4(4\sqrt{3} + \sqrt{6} - 4)$ (C) $8(4\sqrt{3} + 3\sqrt{6} - 9)$ (D) $8(4\sqrt{3} + 2\sqrt{6} - 9)$

Q69. JEE Main 2022 (Online) 25th June Evening Shift

The area of the region enclosed between the parabolas $y^2 = 2x - 1$ and $y^2 = 4x - 3$ is

(A) $\frac{1}{3}$ (B) $\frac{1}{6}$ (C) $\frac{2}{3}$ (D) $\frac{3}{4}$

Q70. JEE Main 2021 (Online) 1st September Evening Shift

The area, enclosed by the curves $y = \sin x + \cos x$ and $y = |\cos x - \sin x|$ and the lines $x = 0, x = \frac{\pi}{2}$, is :

(A) $2\sqrt{2}(\sqrt{2} - 1)$ (B) $2(\sqrt{2} + 1)$ (C) $4(\sqrt{2} - 1)$ (D) $2\sqrt{2}(\sqrt{2} + 1)$

Q71. JEE Main 2021 (Online) 27th August Evening Shift

The area of the region bounded by the parabola $(y - 2)^2 = (x - 1)$, the tangent to it at the point whose ordinate is 3 and the x-axis is :

(A) 9 (B) 10 (C) 4 (D) 6

Q72. JEE Main 2021 (Online) 27th July Evening Shift

The area of the region bounded by $y - x = 2$ and $x^2 = y$ is equal to :

(A) $\frac{16}{3}$ (B) $\frac{2}{3}$ (C) $\frac{9}{2}$ (D) $\frac{4}{3}$

Q73. JEE Main 2021 (Online) 27th July Morning Shift

If the area of the bounded region
$R = \left\{(x, y) : \max\{0, \log_e x\} \leq y \leq 2^x, \frac{1}{2} \leq x \leq 2\right\}$ is ,
$\alpha(\log_e 2)^{-1} + \beta(\log_e 2) + \gamma$, then the value of $(\alpha + \beta - 2\lambda)^2$ is equal to :

(A) 8 (B) 2 (C) 4 (D) 1

Q74. JEE Main 2021 (Online) 25th July Morning Shift

The area (in sq. units) of the region, given by the set $\{(x, y) \in R \times R | x \geq 0, 2x^2 \leq y \leq 4 - 2x\}$ is :

(A) $\frac{8}{3}$ (B) $\frac{17}{3}$ (C) $\frac{13}{3}$ (D) $\frac{7}{3}$

Q75. JEE Main 2021 (Online) 18th March Evening Shift

The area bounded by the curve $4y^2 = x^2(4 - x)(x - 2)$ is equal to :

(A) $\frac{\pi}{16}$ (B) $\frac{\pi}{8}$ (C) $\frac{3\pi}{2}$ (D) $\frac{3\pi}{8}$

Q76. JEE Main 2021 (Online) 26th February Evening Shift

Let A_1 be the area of the region bounded by the curves y = sinx, y = cosx and y-axis in the first quadrant. Also, let A_2 be the area of the region bounded by the curves y = sinx, y = cosx, x-axis and x = $\frac{\pi}{2}$ in the first quadrant. Then,

(A) $A_1 : A_2 = 1 : \sqrt{2}$ and $A_1 + A_2 = 1$ (B) $A_1 = A_2$ and $A_1 + A_2 = \sqrt{2}$

(C) $2A_1 = A_2$ and $A_1 + A_2 = 1 + \sqrt{2}$ (D) $A_1 : A_2 = 1 : 2$ and $A_1 + A_2 = 1$

Q77. JEE Main 2021 (Online) 24th February Evening Shift

The area of the region : $R = \{(x, y) : 5x^2 \le y \le 2x^2 + 9\}$ is :

(A) $6\sqrt{3}$ square units (B) $12\sqrt{3}$ square units

(C) $11\sqrt{3}$ square units (D) $9\sqrt{3}$ square units

Q78. JEE Main 2021 (Online) 24th February Morning Shift

The area (in sq. units) of the part of the circle $x^2 + y^2 = 36$, which is outside the parabola $y^2 = 9x$, is :

(A) $12\pi - 3\sqrt{3}$ (B) $24\pi + 3\sqrt{3}$ (C) $24\pi - 3\sqrt{3}$ (D) $12\pi + 3\sqrt{3}$

Q79. JEE Main 2020 (Online) 6th September Evening Slot

The area (in sq. units) of the region enclosed
by the curves $y = x^2 - 1$ and $y = 1 - x^2$ is equal to :

(A) $\frac{8}{3}$ (B) $\frac{4}{3}$ (C) $\frac{7}{2}$ (D) $\frac{16}{3}$

Q80. JEE Main 2020 (Online) 6th September Morning Slot

The area (in sq. units) of the region
A = {(x, y) : $|x| + |y| \le 1, 2y^2 \ge |x|$}

(A) $\frac{1}{6}$ (B) $\frac{5}{6}$ (C) $\frac{1}{3}$ (D) $\frac{7}{6}$

Q81. JEE Main 2020 (Online) 5th September Evening Slot

The area (in sq. units) of the region

A = {(x, y) : (x − 1)[x] $\leq$ y $\leq 2\sqrt{x}$, 0 $\leq$ x $\leq$ 2}, where [t]

denotes the greatest integer function, is :

(A) $\frac{8}{3}\sqrt{2}-1$ (B) $\frac{4}{3}\sqrt{2}+1$

(C) $\frac{8}{3}\sqrt{2}-\frac{1}{2}$ (D) $\frac{4}{3}\sqrt{2}-\frac{1}{2}$

Q82. JEE Main 2020 (Online) 3rd September Morning Slot

The area (in sq. units) of the region

{ (x, y) : 0 $\leq$ y $\leq x^2 + 1$, 0 $\leq$ y $\leq$ x + 1,

$\frac{1}{2} \leq$ x $\leq$ 2 } is :

(A) $\frac{79}{16}$ (B) $\frac{79}{24}$ (C) $\frac{23}{6}$ (D) $\frac{23}{16}$

Q83. JEE Main 2020 (Online) 2nd September Evening Slot

Consider a region R = {(x, y) $\in$ R : $x^2 \leq$ y $\leq$ 2x}. if a line y = α divides the area of region R into two equal parts, then which of the following is true?

(A) $3\alpha^2 - 8\alpha + 8 = 0$ (B) $\alpha^3 - 6\alpha^{3/2} - 16 = 0$

(C) $3\alpha^2 - 8\alpha^{3/2} + 8 = 0$ (D) $\alpha^3 - 6\alpha^2 + 16 = 0$

Q84. JEE Main 2020 (Online) 2nd September Morning Slot

Area (in sq. units) of the region outside

$\frac{|x|}{2}+\frac{|y|}{3}=1$ and inside the ellipse $\frac{x^2}{4}+\frac{y^2}{9}=1$ is :

(A) $6\,(4-\pi)$ (B) $3\,(4-\pi)$ (C) $6\,(\pi-2)$ (D) $3\,(\pi-2)$

Q85. JEE Main 2020 (Online) 9th January Evening Slot

Given : $f(x) = \begin{cases} x & , \quad 0 \le x < \frac{1}{2} \\ \frac{1}{2} & , \quad x = \frac{1}{2} \\ 1-x & , \quad \frac{1}{2} < x \le 1 \end{cases}$

and $g(x) = \left(x - \frac{1}{2}\right)^2, x \in R$

Then the area (in sq. units) of the region bounded by the curves, y = f(x) and y = g(x) between the lines, 2x = 1 and 2x = $\sqrt{3}$, is :

(A) $\frac{1}{2} + \frac{\sqrt{3}}{4}$ (B) $\frac{1}{2} - \frac{\sqrt{3}}{4}$

(C) $\frac{1}{3} + \frac{\sqrt{3}}{4}$ (D) $\frac{\sqrt{3}}{4} - \frac{1}{3}$

Q86. JEE Main 2020 (Online) 8th January Evening Slot

The area (in sq. units) of the region

$\{(x,y) \in R^2 : x^2 \le y \le 3 - 2x\}$, is :

(A) $\frac{34}{3}$ (B) $\frac{29}{3}$ (C) $\frac{31}{3}$ (D) $\frac{32}{3}$

Q87. JEE Main 2020 (Online) 8th January Morning Slot

For a > 0, let the curves $C_1 : y^2 = ax$ and $C_2 : x^2 = ay$ intersect at origin O and a point P. Let the line x = b (0 < b < a) intersect the chord OP and the x-axis at points Q and R, respectively. If the line x = b bisects the area bounded by the curves, C_1 and C_2, and the area of $\Delta OQR = \frac{1}{2}$, then 'a' satisfies the equation :

(A) $x^6 - 12x^3 + 4 = 0$ (B) $x^6 - 12x^3 - 4 = 0$

(C) $x^6 + 6x^3 - 4 = 0$ (D) $x^6 - 6x^3 + 4 = 0$

Q88. JEE Main 2020 (Online) 7th January Evening Slot

The area (in sq. units) of the region
$\{(x, y) \in R^2 \mid 4x^2 \le y \le 8x + 12\}$ is :

(A) $\frac{125}{3}$ (B) $\frac{128}{3}$ (C) $\frac{127}{3}$ (D) $\frac{124}{3}$

Q89. JEE Main 2020 (Online) 7th January Morning Slot

The area of the region, enclosed by the circle $x^2 + y^2 = 2$ which is not common to the region bounded by the parabola $y^2 = x$ and the straight line y = x, is:

(A) $\frac{1}{6}(24\pi - 1)$ (B) $\frac{1}{3}(12\pi - 1)$

(C) $\frac{1}{3}(6\pi - 1)$ (D) $\frac{1}{6}(12\pi - 1)$

Q90. JEE Main 2019 (Online) 12th April Evening Slot

If the area (in sq. units) bounded by the parabola $y^2 = 4\lambda x$ and the line $y = \lambda x$, $\lambda > 0$, is $\frac{1}{9}$, then λ is equal to :

(A) $4\sqrt{3}$ (B) $2\sqrt{6}$ (C) 48 (D) 24

Q91. JEE Main 2019 (Online) 12th April Morning Slot

If the area (in sq. units) of the region $\{(x, y) : y^2 \leq 4x, x + y \leq 1, x \geq 0, y \geq 0\}$ is $a\sqrt{2} + b$, then a − b is equal to :

(A) $\frac{8}{3}$ (B) $-\frac{2}{3}$ (C) 6 (D) $\frac{10}{3}$

Q92. JEE Main 2019 (Online) 10th April Evening Slot

The area (in sq.units) of the region bounded by the curves $y = 2^x$ and $y = |x + 1|$, in the first quadrant is :

(A) $\frac{1}{2}$ (B) $\frac{3}{2}$
(C) $\frac{3}{2} - \frac{1}{\log_e 2}$ (D) $\log_e 2 + \frac{3}{2}$

Q93. JEE Main 2019 (Online) 9th April Evening Slot

The area (in sq. units) of the region
$A = \{(x, y) : \frac{y^2}{2} \leq x \leq y + 4\}$ is :-

(A) 30 (B) 18 (C) $\frac{53}{3}$ (D) 16

Q94. JEE Main 2019 (Online) 9th April Morning Slot

The area (in sq. units) of the region

$A = \{(x, y) : x^2 \leq y \leq x + 2\}$ is

(A) $\frac{31}{6}$ (B) $\frac{10}{3}$ (C) $\frac{13}{6}$ (D) $\frac{9}{2}$

Q95. JEE Main 2019 (Online) 8th April Evening Slot

Let $S(\alpha) = \{(x, y) : y^2 \leq x, 0 \leq x \leq \alpha\}$ and $A(\alpha)$ is area of the region $S(\alpha)$. If for a λ, $0 < \lambda < 4$, $A(\lambda) : A(4) = 2 : 5$, then λ equals

(A) $2\left(\frac{4}{25}\right)^{\frac{1}{3}}$ (B) $2\left(\frac{2}{5}\right)^{\frac{1}{3}}$
(C) $4\left(\frac{4}{25}\right)^{\frac{1}{3}}$ (D) $4\left(\frac{2}{5}\right)^{\frac{1}{3}}$

Q96. JEE Main 2019 (Online) 8th April Morning Slot

The area (in sq. units) of the region
$A = \{(x, y) \in R \times R | \ 0 \leq x \leq 3, 0 \leq y \leq 4, y \leq x^2 + 3x\}$ is :

(A) $\frac{59}{6}$ (B) $\frac{26}{3}$ (C) 8 (D) $\frac{53}{6}$

Q97. JEE Main 2019 (Online) 12th January Morning Slot

The area (in sq. units) of the region bounded by the parabola, $y = x^2 + 2$ and the lines, $y = x + 1$, $x = 0$ and $x = 3$, is

(A) $\frac{15}{4}$ (B) $\frac{15}{2}$ (C) $\frac{21}{2}$ (D) $\frac{17}{4}$

Q98. JEE Main 2019 (Online) 11th January Evening Slot

The area (in sq. units) in the first quadrant bounded by the parabola, $y = x^2 + 1$, the tangent to it at the point (2, 5) and the coordinate axes is :

(A) $\frac{8}{3}$ (B) $\frac{14}{3}$ (C) $\frac{187}{24}$ (D) $\frac{37}{24}$

Q99. JEE Main 2019 (Online) 11th January Morning Slot

The area (in sq. units) of the region bounded by the curve $x^2 = 4y$ and the straight line $x = 4y - 2$ is :

(A) $\frac{3}{4}$ (B) $\frac{5}{4}$ (C) $\frac{7}{8}$ (D) $\frac{9}{8}$

Q100. JEE Main 2019 (Online) 10th January Morning Slot

If the area enclosed between the curves $y = kx^2$ and $x = ky^2$, $(k > 0)$, is 1 square unit. Then k is -

(A) $\sqrt{3}$ (B) $\frac{\sqrt{3}}{2}$ (C) $\frac{2}{\sqrt{3}}$ (D) $\frac{1}{\sqrt{3}}$

Q101. JEE Main 2019 (Online) 9th January Evening Slot

The area of the region

$A = \{(x, y) : 0 \leq y \leq x|x| + 1 \text{ and } -1 \leq x \leq 1\}$ in sq. units, is :

(A) $\frac{2}{3}$ (B) 2 (C) $\frac{4}{3}$ (D) $\frac{1}{3}$

Q102. JEE Main 2019 (Online) 9th January Morning Slot

The area (in sq. units) bounded by the parabola $y = x^2 - 1$, the tangent at the point (2, 3) to it and the y-axis is :

(A) $\frac{56}{3}$ (B) $\frac{32}{3}$ (C) $\frac{8}{3}$ (D) $\frac{14}{3}$

Q103. JEE Main 2018 (Online) 16th April Morning Slot

If the area of the region bounded by the curves, $y = x^2, y = \frac{1}{x}$ and the lines y = 0 and x= t (t >1) is 1 sq. unit, then t is equal to :

(A) $e^{\frac{3}{2}}$ (B) $\frac{4}{3}$ (C) $\frac{3}{2}$ (D) $e^{\frac{2}{3}}$

Q104. JEE Main 2018 (Offline)

Let g(x) = $\cos x^2$, f(x) = $\sqrt{x}$ and $\alpha, \beta\,(\alpha < \beta)$ be the roots of the quadratic equation $18x^2 - 9\pi x + \pi^2 = 0$. Then the area (in sq. units) bounded by the curve
y = (gof)(x) and the lines $x = \alpha, x = \beta$ and y = 0 is :

(A) $\frac{1}{2}\left(\sqrt{2}-1\right)$ (B) $\frac{1}{2}\left(\sqrt{3}-1\right)$

(C) $\frac{1}{2}\left(\sqrt{3}+1\right)$ (D) $\frac{1}{2}\left(\sqrt{3}-\sqrt{2}\right)$

Q105. JEE Main 2018 (Online) 15th April Morning Slot

The area (in sq. units) of the region

$\{x \in \mathbf{R} : x \geq 0, y \geq 0, y \geq x - 2 \text{ and } y \leq \sqrt{x}\}$, is :

(A) $\frac{13}{3}$ (B) $\frac{8}{3}$ (C) $\frac{10}{3}$ (D) $\frac{5}{3}$

Q106. JEE Main 2017 (Online) 8th April Morning Slot

The area (in sq. units) of the smaller portion enclosed between the curves, $x^2 + y^2 = 4$ and $y^2 = 3x$, is :

(A) $\frac{1}{2\sqrt{3}} + \frac{\pi}{3}$ (B) $\frac{1}{\sqrt{3}} + \frac{2\pi}{3}$

(C) $\frac{1}{2\sqrt{3}} + \frac{2\pi}{3}$ (D) $\frac{1}{\sqrt{3}} + \frac{4\pi}{3}$

Q107. JEE Main 2017 (Offline)

The area (in sq. units) of the region

$\{(x, y) : x \geq 0, x + y \leq 3, x^2 \leq 4y \text{ and } y \leq 1 + \sqrt{x}\}$ is

(A) $\frac{3}{2}$ (B) $\frac{7}{3}$ (C) $\frac{5}{2}$ (D) $\frac{59}{12}$

Q108. JEE Main 2016 (Online) 9th April Morning Slot

The area (in sq. units) of the region described by

A= {(x, y) |y$\geq x^2 - 5x + 4, x + y \geq 1, y \leq 0$} is :

(A) $\frac{7}{2}$ (B) $\frac{19}{6}$ (C) $\frac{13}{6}$ (D) $\frac{17}{6}$

Q109. JEE Main 2016 (Offline)

The area (in sq. units) of the region $\{(x, y) : y^2 \geq 2x \text{ and } x^2 + y^2 \leq 4x, x \geq 0, y \geq 0\}$ is :

(A) $\pi - \frac{4\sqrt{2}}{3}$ (B) $\frac{\pi}{2} - \frac{2\sqrt{2}}{3}$

(C) $\pi - \frac{4}{3}$ (D) $\pi - \frac{8}{3}$

Q110. JEE Main 2015 (Offline)

The area (in sq. units) of the region described by

$\{(x, y) : y^2 \leq 2x$ and $y \geq 4x - 1\}$ is :

(A) $\frac{15}{64}$ (B) $\frac{9}{32}$ (C) $\frac{7}{32}$ (D) $\frac{5}{64}$

Q111. JEE Main 2014 (Offline)

The area of the region described by
$A = \{(x, y) : x^2 + y^2 \leq 1$ and $y^2 \leq 1 - x\}$ is :

(A) $\frac{\pi}{2} - \frac{2}{3}$ (B) $\frac{\pi}{2} + \frac{2}{3}$ (C) $\frac{\pi}{2} + \frac{4}{3}$ (D) $\frac{\pi}{2} - \frac{4}{3}$

Q112. JEE Main 2013 (Offline)

The area (in square units) bounded by the curves $y = \sqrt{x}$, $2y - x + 3 = 0$, x-axis, and lying in the first quadrant is :

(A) 9 (B) 36 (C) 18 (D) $\frac{27}{4}$

Q113. AIEEE 2012

The area between the parabolas $x^2 = \frac{y}{4}$ and $x^2 = 9y$ and the straight line $y = 2$ is :

(A) $20\sqrt{2}$ (B) $\frac{10\sqrt{2}}{3}$ (C) $\frac{20\sqrt{2}}{3}$ (D) $10\sqrt{2}$

Q114. AIEEE 2011

The area of the region enclosed by the curves $y = x, x = e, y = \frac{1}{x}$ and the positive x-axis is :

(A) 1 square unit (B) $\frac{3}{2}$ square units

(C) $\frac{5}{2}$ square units (D) $\frac{1}{2}$ square unit

Q115. AIEEE 2010

The area bounded by the curves $y = \cos x$ and $y = \sin x$ between the ordinates $x = 0$ and $x = \frac{3\pi}{2}$ is

(A) $4\sqrt{2} + 2$ (B) $4\sqrt{2} - 1$ (C) $4\sqrt{2} + 1$ (D) $4\sqrt{2} - 2$

Q116. AIEEE 2009

The area of the region bounded by the parabola $(y - 2)^2 = x - 1$, the tangent of the parabola at the point $(2, 3)$ and the x-axis is :

(A) 6 (B) 9 (C) 12 (D) 3

Q117. AIEEE 2008

The area of the plane region bounded by the curves $x + 2y^2 = 0$ and $x + 3y^2 = 1$ is equal to :

(A) $\frac{5}{3}$ (B) $\frac{1}{3}$ (C) $\frac{2}{3}$ (D) $\frac{4}{3}$

Q118. AIEEE 2007

The area enclosed between the curves $y^2 = x$ and $y = |x|$ is :

(A) $1/6$ (B) $1/3$ (C) $2/3$ (D) 1

Q119. AIEEE 2005

The area enclosed between the curve $y = \log_e(x + e)$ and the coordinate axes is :

(A) 1 (B) 2 (C) 3 (D) 4

Q120. AIEEE 2005

The parabolas $y^2 = 4x$ and $x^2 = 4y$ divide the square region bounded by the lines $x = 4, y = 4$ and the coordinate axes. If S_1, S_2, S_3 are respectively the areas of these parts numbered from top to bottom ; then S_1, S_2, S_3 is :

(A) $1 : 2 : 1$ (B) $1 : 2 : 3$ (C) $2 : 1 : 2$ (D) $1 : 1 : 1$

Q121. AIEEE 2005

Let $f(x)$ be a non - negative continuous function such that the area bounded by the curve $y = f(x)$, x-axis and the ordinates $x = \frac{\pi}{4}$ and $x = \beta > \frac{\pi}{4}$ is $\left(\beta \sin\beta + \frac{\pi}{4}\cos\beta + \sqrt{2}\beta\right)$. Then $f\left(\frac{\pi}{2}\right)$ is

(A) $\left(\frac{\pi}{4} + \sqrt{2} - 1\right)$ (B) $\left(\frac{\pi}{4} - \sqrt{2} + 1\right)$

(C) $\left(1 - \frac{\pi}{4} - \sqrt{2}\right)$ (D) $\left(1 - \frac{\pi}{4} + \sqrt{2}\right)$

Q122. AIEEE 2004

The area of the region bounded by the curves
$y = |x - 2|, x = 1, x = 3$ and the x-axis is :

(A) 4 (B) 2 (C) 3 (D) 1

Q123. AIEEE 2003

The area of the region bounded by the curves $y = |x - 1|$ and $y = 3 - |x|$ is :

(A) 6 sq. units (B) 2 sq. units (C) 3 sq. units (D) 4 sq. units

Q124. AIEEE 2002

The area bounded by the curves $y = \ln x, y = \ln |x|, y = |\ln x|$ and $y = |\ln |x||$ is :

(A) 4sq. units (B) 6sq. units (C) 10sq. units (D) none of these

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Q125. JEE Main 2026 (Online) 5th April Evening Shift

Let $f : \mathbf{R} \to \mathbf{R}$ be a function such that $f(x) + 3f\left(\frac{\pi}{2} - x\right) = \sin x, x \in \mathbf{R}$. Let the maximum value of f on $\mathbf{R}$ be α. If the area of the region bounded by the curves $g(x) = x^2$ and $h(x) = \beta x^3, \beta > 0$, is α^2, then $30\beta^3$ is equal to ____ .

Q126. JEE Main 2026 (Online) 2nd April Evening Shift

If the area of the region bounded by $16x^2 - 9y^2 = 144$ and $8x - 3y = 24$ is A, then $3(A + 6\log_e(3))$ is equal to _____.

Q127. JEE Main 2026 (Online) 23rd January Morning Shift

Let the area of the region bounded by the curve $y = \max\{\sin x, \cos x\}$, lines $x = 0, x = \frac{3\pi}{2}$, and the x-axis be A . Then, $\mathrm{A} + \mathrm{A}^2$ is equal to ____.

Q128. JEE Main 2025 (Online) 8th April Evening Shift

Let the area of the bounded region $\{(x, y) : 0 \le 9x \le y^2, y \ge 3x - 6\}$ be A. Then $6A$ is equal to _____.

Q129. JEE Main 2025 (Online) 4th April Morning Shift

If the area of the region $\{(x, y) : |x - 5| \le y \le 4\sqrt{x}\}$ is A, then $3A$ is equal to _____.

Q130. JEE Main 2025 (Online) 3rd April Morning Shift

The area of the region bounded by the curve $y = \max\{|x|, x|x - 2|\}$, the x-axis and the lines $x = -2$ and $x = 4$ is equal to_____

Q131. JEE Main 2025 (Online) 2nd April Morning Shift

If the area of the region $\left\{(x, y):\left|4-x^2\right| \leq y \leq x^2, y \leq 4, x \geq 0\right\}$ is $\left(\frac{80 \sqrt{2}}{\alpha}-\beta\right), \alpha, \beta \in \mathbf{N}$, then $\alpha+\beta$ is equal to _____.

Q132. JEE Main 2025 (Online) 23rd January Morning Shift

If the area of the larger portion bounded between the curves $x^2+y^2=25$ and $\mathrm{y}=|\mathrm{x}-1|$ is $\frac{1}{4}(\mathrm{b} \pi+\mathrm{c}), \mathrm{b}, \mathrm{c} \in N$, then $\mathrm{b}+\mathrm{c}$ is equal to _____

Q133. JEE Main 2024 (Online) 8th April Morning Shift

Let the area of the region enclosed by the curve $y=\min \{\sin x, \cos x\}$ and the x axis between $x=-\pi$ to $x=\pi$ be A. Then A^2 is equal to _____.

Q134. JEE Main 2024 (Online) 5th April Morning Shift

The area of the region enclosed by the parabolas $y=x^2-5x$ and $y=7x-x^2$ is _____.

Q135. JEE Main 2024 (Online) 1st February Evening Shift

Three points $\mathrm{O}(0,0), \mathrm{P}\left(\mathrm{a}, \mathrm{a}^2\right), \mathrm{Q}\left(-\mathrm{b}, \mathrm{b}^2\right), \mathrm{a}>0,\ \mathrm{b}>0$, are on the parabola $y=x^2$. Let S_1 be the area of the region bounded by the line PQ and the parabola, and S_2 be the area of the triangle OPQ. If the minimum value of $\frac{\mathrm{S}_1}{\mathrm{S}_2}$ is $\frac{\mathrm{m}}{\mathrm{n}}, \gcd(\mathrm{m}, \mathrm{n})=1$, then $\mathrm{m}+\mathrm{n}$ is equal to _____.

Q136. JEE Main 2024 (Online) 1st February Evening Shift

The sum of squares of all possible values of k, for which area of the region bounded by the parabolas $2y^2=\mathrm{k}x$ and $\mathrm{k}y^2=2(y-x)$ is maximum, is equal to :

Q137. JEE Main 2024 (Online) 30th January Evening Shift

The area of the region enclosed by the parabola $(y-2)^2=x-1$, the line $x-2y+4=0$ and the positive coordinate axes is _____.

Q138. JEE Main 2024 (Online) 29th January Evening Shift

Let the area of the region $\left\{(x, y): 0 \leq x \leq 3, 0 \leq y \leq \min \left\{x^2+2, 2x+2\right\}\right\}$ be A. Then $12\ \mathrm{A}$ is equal to _____.

Q139. JEE Main 2024 (Online) 29th January Morning Shift

The area (in sq. units) of the part of the circle $x^2+y^2=169$ which is below the line $5x-y=13$ is $\frac{\pi\alpha}{2\beta}-\frac{65}{2}+\frac{\alpha}{\beta}\sin^{-1}\left(\frac{12}{13}\right)$, where α, β are coprime numbers. Then $\alpha+\beta$ is equal to _____.

Q140. JEE Main 2024 (Online) 29th January Morning Shift

If the points of intersection of two distinct conics $x^2+y^2=4b$ and $\frac{x^2}{16}+\frac{y^2}{b^2}=1$ lie on the curve $y^2=3x^2$, then $3\sqrt{3}$ times the area of the rectangle formed by the intersection points is _____.

Q141. JEE Main 2024 (Online) 27th January Evening Shift

If the area of the region $\left\{(x, y): 0 \leq y \leq \min\left\{2x, 6x-x^2\right\}\right\}$ is A, then $12\,\mathrm{A}$ is equal to _____.

Q142. JEE Main 2024 (Online) 27th January Morning Shift

Let the area of the region $\left\{(x, y): x-2y+4 \geqslant 0, x+2y^2 \geqslant 0, x+4y^2 \leq 8, y \geqslant 0\right\}$ be $\frac{\mathrm{m}}{\mathrm{n}}$, where m and n are coprime numbers. Then $\mathrm{m}+\mathrm{n}$ is equal to _____.

Q143. JEE Main 2023 (Online) 15th April Morning Shift

If the area bounded by the curve $2y^2=3x$, lines $x+y=3, y=0$ and outside the circle $(x-3)^2+y^2=2$ is A, then $4(\pi+4A)$ is equal to _____.

Q144. JEE Main 2023 (Online) 11th April Evening Shift

If A is the area in the first quadrant enclosed by the curve $\mathrm{C}: 2\mathrm{x}^2-\mathrm{y}+1=0$, the tangent to C at the point $(1,3)$ and the line $\mathrm{x}+\mathrm{y}=1$, then the value of $60\,\mathrm{A}$ is _____.

Q145. JEE Main 2023 (Online) 10th April Evening Shift

If the area of the region $\left\{(x, \mathrm{y}):\left|x^2-2\right| \leq y \leq x\right\}$ is A, then $6\mathrm{A}+16\sqrt{2}$ is equal to _____.

Q146. JEE Main 2023 (Online) 10th April Morning Shift

Let $y=p(x)$ be the parabola passing through the points $(-1,0),(0,1)$ and $(1,0)$. If the area of the region $\{(x, y):(x+1)^2+(y-1)^2 \leq 1, y \leq p(x)\}$ is A, then $12(\pi-4A)$ is equal to _____.

Q147. JEE Main 2023 (Online) 8th April Evening Shift

Let the area enclosed by the lines $x+y=2, \mathrm{y}=0, x=0$ and the curve $f(x)=\min\left\{x^2+\frac{3}{4}, 1+[x]\right\}$ where $[x]$ denotes the greatest integer $\leq x$, be A. Then the value of $12\ \mathrm{A}$ is _____.

Q148. JEE Main 2023 (Online) 6th April Morning Shift

If the area of the region $S=\left\{(x, y): 2y-y^2 \leq x^2 \leq 2y, x \geq y\right\}$ is equal to $\frac{n+2}{n+1}-\frac{\pi}{n-1}$, then the natural number n is equal to _____.

Q149. JEE Main 2023 (Online) 1st February Morning Shift

Let A be the area bounded by the curve $y=x|x-3|$, the x-axis and the ordinates $x=-1$ and $x=2$. Then $12A$ is equal to _____.

Q150. JEE Main 2023 (Online) 31st January Evening Shift

Let the area of the region

$\left\{(x, y):|2x-1| \leq y \leq \left|x^2-x\right|, 0 \leq x \leq 1\right\}$ be A.

Then $(6\ \mathrm{A}+11)^2$ is equal to

Q151. JEE Main 2023 (Online) 31st January Morning Shift

Let for $x \in \mathbb{R}$,

$f(x)=\frac{x+|x|}{2}$ and $g(x)=\begin{cases}x, & x<0 \\ x^2, & x \geq 0\end{cases}$.

Then area bounded by the curve $y=(f \circ g)(x)$ and the lines $y=0, 2y-x=15$ is equal to _____.

Q152. JEE Main 2023 (Online) 30th January Evening Shift

Let A be the area of the region

$\left\{(x, y): y \geq x^2, y \geq (1-x)^2, y \leq 2x(1-x)\right\}$.

Then $540\ \mathrm{A}$ is equal to :

Q153. JEE Main 2023 (Online) 30th January Morning Shift

Let α be the area of the larger region bounded by the curve $y^2=8x$ and the lines $y=x$ and $x=2$, which lies in the first quadrant. Then the value of 3α is equal to _____.

Q154. JEE Main 2023 (Online) 25th January Morning Shift

If the area enclosed by the parabolas $\mathrm{P_1 : 2y = 5x^2}$ and $\mathrm{P_2 : x^2 - y + 6 = 0}$ is equal to the area enclosed by $\mathrm{P_1}$ and $\mathrm{y = \alpha x, \alpha > 0}$, then α^3 is equal to _____.

Q155. JEE Main 2023 (Online) 24th January Evening Shift

If the area of the region bounded by the curves $y^2 - 2y = -x, x + y = 0$ is A, then 8 A is equal to _____

Q156. JEE Main 2022 (Online) 25th July Evening Shift

Let the area enclosed by the x-axis, and the tangent and normal drawn to the curve $4x^3 - 3xy^2 + 6x^2 - 5xy - 8y^2 + 9x + 14 = 0$ at the point (−2, 3) be A. Then 8A is equal to _____.

Q157. JEE Main 2022 (Online) 30th June Morning Shift

If for some α > 0, the area of the region $\{(x, y) : |x + \alpha| \le y \le 2 - |x|\}$ is equal to $\frac{3}{2}$, then the area of the region $\{(x, y) : 0 \le y \le x + 2\alpha,\ |x| \le 1\}$ is equal to _____.

Q158. JEE Main 2022 (Online) 29th June Evening Shift

For real numbers a, b (a > b > 0), let

Area $\left\{(x, y) : x^2 + y^2 \le a^2 \text{ and } \frac{x^2}{a^2} + \frac{y^2}{b^2} \ge 1\right\} = 30\pi$

and

Area $\left\{(x, y) : x^2 + y^2 \ge b^2 \text{ and } \frac{x^2}{a^2} + \frac{y^2}{b^2} \le 1\right\} = 18\pi$

Then, the value of $(a - b)^2$ is equal to _____.

Q159. JEE Main 2022 (Online) 27th June Evening Shift

If the area of the region $\left\{(x, y) : x^{\frac{2}{3}} + y^{\frac{2}{3}} \le 1,\ x + y \ge 0,\ y \ge 0\right\}$ is A, then $\frac{256A}{\pi}$ is equal to _____.

Q160. JEE Main 2022 (Online) 27th June Morning Shift

Let

$A_1 = \{(x, y) : |x| \le y^2, |x| + 2y \le 8\}$ and

$A_2 = \{(x, y) : |x| + |y| \le k\}$. If 27 (Area A_1) = 5 (Area A_2), then k is equal to :

Q161. JEE Main 2022 (Online) 24th June Evening Shift

The area (in sq. units) of the region enclosed between the parabola $y^2 = 2x$ and the line $x + y = 4$ is _____.

Q162. JEE Main 2022 (Online) 24th June Morning Shift

Let S be the region bounded by the curves $y = x^3$ and $y^2 = x$. The curve $y = 2|x|$ divides S into two regions of areas R_1, R_2. If max $\{R_1, R_2\} = R_2$, then $\frac{R_2}{R_1}$ is equal to _____.

Q163. JEE Main 2021 (Online) 31st August Evening Shift

If the line $y = mx$ bisects the area enclosed by the lines $x = 0$, $y = 0$, $x = \frac{3}{2}$ and the curve $y = 1 + 4x - x^2$, then 12 m is equal to _____.

Q164. JEE Main 2021 (Online) 26th August Evening Shift

Let a and b respectively be the points of local maximum and local minimum of the function $f(x) = 2x^3 - 3x^2 - 12x$. If A is the total area of the region bounded by $y = f(x)$, the x-axis and the lines $x = a$ and $x = b$, then 4A is equal to _____.

Q165. JEE Main 2021 (Online) 26th August Morning Shift

The area of the region $S = \{(x, y) : 3x^2 \le 4y \le 6x + 24\}$ is _____.

Q166. JEE Main 2021 (Online) 22th July Evening Shift

The area (in sq. units) of the region bounded by the curves $x^2 + 2y - 1 = 0$, $y^2 + 4x - 4 = 0$ and $y^2 - 4x - 4 = 0$, in the upper half plane is _____.

Q167. JEE Main 2021 (Online) 20th July Morning Shift

Let T be the tangent to the ellipse E : $x^2 + 4y^2 = 5$ at the point P(1, 1). If the area of the region bounded by the tangent T, ellipse E, lines $x = 1$ and $x = \sqrt{5}$ is $\alpha\sqrt{5} + \beta + \gamma \cos^{-1}\left(\frac{1}{\sqrt{5}}\right)$, then $|\alpha + \beta + \gamma|$ is equal to _____.

Q168. JEE Main 2021 (Online) 17th March Evening Shift

Let f : $[-3, 1] \to$ R be given as

$$f(x) = \begin{cases} \min\{(x+6), x^2\}, -3 \le x \le 0 \\ \max\{\sqrt{x}, x^2\}, \ 0 \le x \le 1. \end{cases}$$

If the area bounded by y = f(x) and x-axis is A, then the value of 6A is equal to _____.

Q169. JEE Main 2021 (Online) 26th February Morning Shift

The area bounded by the curve y = || x $-$ 1 | $-$ 2 | and the line y = 2 is _____.

Q170. JEE Main 2021 (Online) 25th February Morning Shift

The graphs of sine and cosine functions, intersect each other at a number of points and between two consecutive points of intersection, the two graphs enclose the same area A. Then A^4 is equal to _____.

Answer Key

MCQ (Single Correct Answer)

1. **A**	2. **A**	3. **B**	4. **A**	5. **B**
6. **C**	7. **B**	8. **A**	9. **D**	10. **D**
11. **A**	12. **A**	13. **A**	14. **A**	15. **A**
16. **C**	17. **A**	18. **B**	19. **D**	20. **D**
21. **C**	22. **A**	23. **B**	24. **B**	25. **C**
26. **C**	27. **B**	28. **A**	29. **D**	30. **B**
31. **A**	32. **C**	33. **D**	34. **C**	35. **C**
36. **A**	37. **A**	38. **C**	39. **D**	40. **A**
41. **C**	42. **C**	43. **B**	44. **C**	45. **D**
46. **C**	47. **B**	48. **A**	49. **B**	50. **C**
51. **C**	52. **B**	53. **C**	54. **B**	55. **D**
56. **B**	57. **D**	58. **C**	59. **C**	60. **D**
61. **B**	62. **B**	63. **C**	64. **C**	65. **C**
66. **B**	67. **C**	68. **D**	69. **A**	70. **A**
71. **A**	72. **C**	73. **B**	74. **D**	75. **C**
76. **A**	77. **B**	78. **C**	79. **A**	80. **B**
81. **C**	82. **B**	83. **C**	84. **C**	85. **D**
86. **D**	87. **A**	88. **B**	89. **D**	90. **D**
91. **C**	92. **C**	93. **B**	94. **D**	95. **C**
96. **A**	97. **B**	98. **D**	99. **D**	100. **D**
101. **B**	102. **C**	103. **D**	104. **B**	105. **C**
106. **D**	107. **C**	108. **B**	109. **D**	110. **B**
111. **C**	112. **A**	113. **C**	114. **B**	115. **D**
116. **B**	117. **D**	118. **A**	119. **A**	120. **D**
121. **D**	122. **D**	123. **D**	124. **A**	

Numerical

125. **16**	126. **24**	127. **12**	128. **15**	129. **368**
130. **12**	131. **22**	132. **77**	133. **16**	134. **198**
135. **7**	136. **8**	137. **5**	138. **164**	139. **171**
140. **432**	141. **304**	142. **119**	143. **42**	144. **16**

145. **27**	146. **16**	147. **17**	148. **5**	149. **62**
150. **125**	151. **72**	152. **25**	153. **22**	154. **600**
155. **36**	156. **170**	157. **4**	158. **12**	159. **36**
160. **6**	161. **18**	162. **19**	163. **26**	164. **114**
165. **27**	166. **2**	167. **1.25**	168. **41**	169. **8**
170. **64**				